

PALMER STATION, Antarctica

WMO: 890610

Lat: 64.774S

Lon: 64.053W

Elev: 26

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 04-19

WBAN: 89061

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-0.4	5.1	-5.1	4.2	0.5	0.3	5.6	7.1	46.6	30.1	42.2	30.1	10.5	240	0.959

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	42.2	38.1	41.0	37.3	40.0	36.5	39.0	41.2	38.1	40.1	37.3	39.3	14.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
37.3	32.7	38.9	36.2	31.3	38.2	35.3	30.2	37.5	15.0	41.5	14.5	40.4	14.2	39.5	45.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
39.6	33.8	28.9	-1.7	45.6	5.5	1.5	-5.7	46.7	-8.9	47.6	-12.0	48.4	-16.0	49.5		
			-1.3	41.2	5.7	1.4	-5.4	42.2	-8.7	43.0	-11.8	43.8	-15.9	44.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.0	35.5	35.3	34.0	30.3	28.8	25.6	22.5	21.2	23.8	27.6	30.7	33.7		
	DBStd	7.03	2.39	2.33	3.01	4.03	4.45	5.06	6.82	7.95	8.11	5.88	4.03	2.77		
	HDD50	7647	451	411	496	591	658	732	851	892	786	694	578	507		
	HDD65	13122	916	831	961	1041	1123	1182	1316	1357	1236	1159	1028	972		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	10.1	7.1	8.7	10.1	10.8	11.3	11.3	10.8	10.8	11.9	11.3	10.1	7.6		
	PrecAvg	36.6	3.0	3.0	4.1	3.5	3.4	2.8	2.7	2.4	2.7	3.5	3.0	2.6		
Precipitation	PrecMax	49.1	4.5	5.1	6.1	5.7	6.3	4.7	4.8	5.4	4.5	5.7	5.1	4.5		
	PrecMin	26.5	1.9	1.8	1.8	2.1	1.1	1.5	1.1	0.8	0.9	2.0	1.4	1.1		
	PrecStd	5.5	0.6	0.8	1.1	1.0	1.3	0.9	0.9	1.0	0.8	0.9	0.9	0.9		
	PrecStd	5.5	0.6	0.8	1.1	1.0	1.3	0.9	0.9	1.0	0.8	0.9	0.9	0.9		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	43.3	43.7	41.1	40.3	39.1	37.1	37.6	38.4	39.8	41.3	43.1		
		MCWB	38.7	39.3	40.1	38.2	37.1	35.8	34.0	34.8	35.2	36.2	37.2	37.7		
	2%	DB	41.9	41.7	41.2	39.2	38.4	36.5	34.7	35.6	36.3	38.0	39.1	40.8		
		MCWB	37.6	37.9	38.0	36.9	35.7	34.1	32.7	33.3	33.8	35.2	35.8	36.2		
	5%	DB	40.6	40.2	39.7	37.5	36.7	34.4	33.5	33.7	34.6	36.5	37.4	39.4		
		MCWB	36.8	36.9	36.9	35.3	34.4	32.5	31.8	31.6	32.7	34.1	34.6	35.4		
	10%	DB	39.4	39.1	38.4	35.9	35.0	32.5	31.9	32.1	33.3	35.1	36.0	38.1		
		MCWB	35.8	36.1	36.1	34.0	33.0	31.1	30.4	30.3	31.9	33.0	33.4	34.4		
	0.4%	WB	40.0	40.7	40.3	38.7	37.8	36.0	34.9	35.3	35.8	37.1	38.5	39.1		
		MCDB	42.6	42.5	42.8	40.3	39.8	37.8	35.9	36.7	38.0	39.0	40.3	41.9		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.5	38.6	38.6	37.1	36.0	34.2	33.2	33.6	34.0	35.7	36.4	37.1		
		MCDB	40.9	40.8	40.5	38.7	37.8	36.1	34.5	35.3	35.4	37.5	38.4	39.6		
	5%	WB	37.5	37.6	37.5	35.8	34.6	32.7	32.0	32.0	33.1	34.5	35.0	36.0		
		MCDB	39.9	39.5	39.2	37.2	36.3	34.2	33.3	33.5	34.3	36.2	36.9	38.4		
	10%	WB	36.5	36.7	36.4	34.2	33.1	31.3	30.5	30.3	32.0	33.3	33.8	35.1		
		MCDB	38.8	38.5	38.1	35.6	34.7	32.5	31.8	31.9	33.2	34.9	35.6	37.3		
	5% DB	MDBR	4.8	4.7	4.7	4.5	4.9	5.3	7.0	8.3	8.8	7.4	6.2	5.2		
		MCDBR	6.7	6.9	6.5	6.5	6.7	6.5	7.3	6.9	8.7	7.5	7.9	7.4		
		MCWBR	4.8	4.8	5.0	5.5	5.7	5.4	6.6	5.9	7.9	5.9	5.9	5.0		
		MCDBR	6.1	6.1	6.3	6.5	6.4	6.5	7.0	8.0	9.2	7.2	7.1	6.5		
	5% WB	MCWBR	4.7	4.7	5.1	5.8	5.7	5.6	6.7	7.4	8.8	6.0	5.6	4.8		
		MCWBR	4.7	4.7	5.1	5.8	5.7	5.6	6.7	7.4	8.8	6.0	5.6	4.8		
Clear-Sky Solar Irradiance	taub	0.290	0.287	0.275	0.248	0.360	N/A	N/A	0.242	0.264	0.284	0.286	0.290			
	taud	2.389	2.409	2.417	2.311	2.885	N/A	N/A	2.165	2.276	2.274	2.326	2.356			
	Ebn at Noon	309	293	266	209	115	N/A	N/A	200	262	289	308	313			
	Edh at Noon	33	29	23	15	7	N/A	N/A	16	25	33	35	36			
All-Sky Solar Radiation	RadAvg	1828	1259	670	245	55	11	34	171	563	1182	1783	2107			
	RadStd	104	84	48	19	3	0	2	14	48	80	138	184			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page